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Mapping the Mind: EEG-MEG Network Classification via Graph Attention Fusion

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1.BACKGROUND

Challenges

- Difficult to fuse EEG and MEG data
- Large difference in channal numbers(70 vs 306)
- Different physical properties and complex spatio-temporal interaction

Our Solution

- Dual-path parallel framework DAF-Net: Deep Attention Fusion Network- TDA: Topological Data Analysis
- Leverage the complementarity of bothmodalties

2.DATASET

Wakeman-Henson Face Recognition Dataset

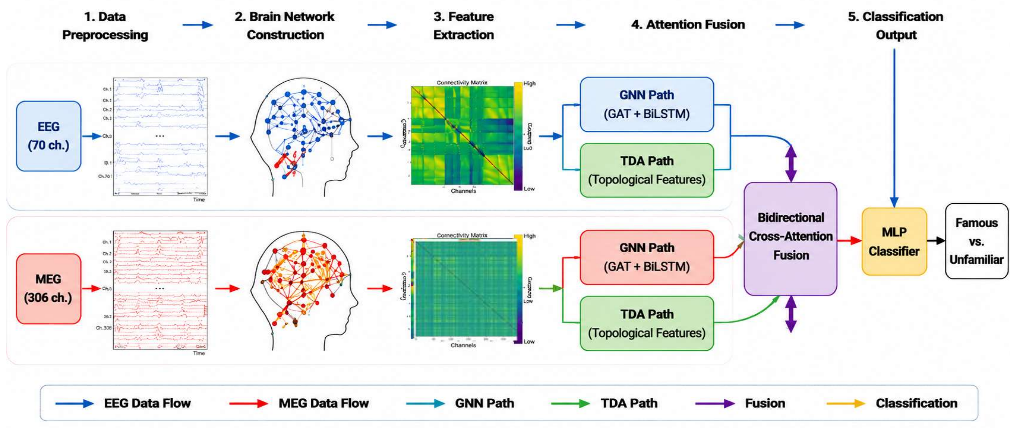
Famous Face | Famous Face | Unfamiliar Face

Trial Start | Famous/Unfamiliar Face | Trial End

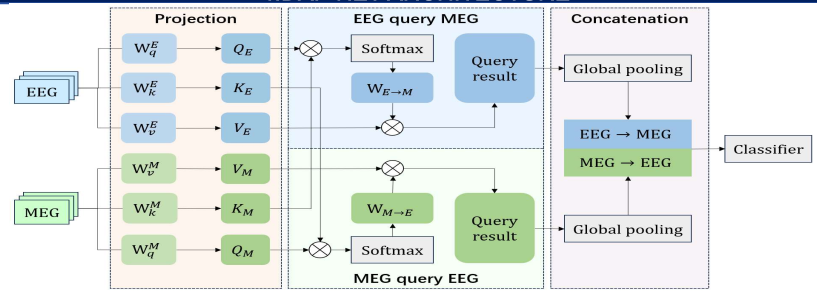
Dataset Description: this dataset has been extensively utilized to investigate neural responses to faces, particularly distinguishing between famous and unfamiliar ones.

- Simultaneous EEG-MEG acquisition
- Binary classification (Famous vs Unfamiliar)
- Total samples: 1,179 trials
  - Famous: 588
  - Unfamiliar: 591
- Data from: sub-002 ...
- Sampling rate: 1100 Hz

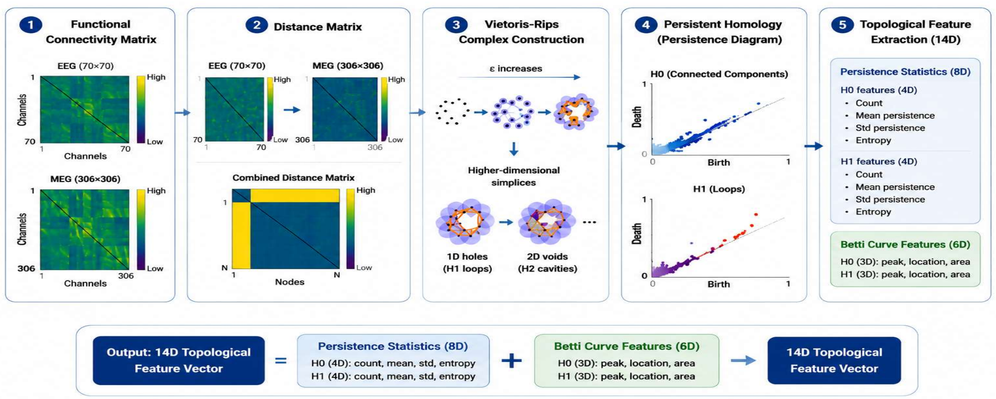
3.METHODOLOGY OVERVIEW: DAF-NET + DUAL-PATH FRAMEWORK



4.DAF-NET ARCHITECTURE



5.TOPOLOGICAL DATA ANALYSIS



6.Main Results

Classification Performance

Method	Data Setting	ACC (Mean ± Std)
Baseline	EEG Only	0.4943 ± 0.0208
	MEG Only	0.4887 ± 0.0138
	Fused EEG-MEG	0.5205 ± 0.0113
GNN	EEG Only	0.5620 ± 0.0308
	MEG Only	0.7643 ± 0.0218
	Fused EEG-MEG	0.7939 ± 0.0301
TDA	EEG Only	0.7470 ± 0.0195
	MEG Only	0.7437 ± 0.0187
	Fused EEG-MEG	0.7689 ± 0.0177

Key Findings

- ✅ Multi-modal fusion outperforms single-modal approaches
- ✅ DAF-Net achieves peak accury of 84%
- ✅ TDA performs well in the 75-78% range

7.Performance Comparison

Advantages of DAF-Net

- 2-14% improvement over single-modal methods
- 5-10% improvement over traditional fusion methods
- Bidrectional attention mechanism is effective
- Fully utililzes spatio-temporal features

Characteristics of TDA

- Strong noise robustense
- Captures global topological structures
- Robust to outliers
- Complements GNN local features

8.Conclusion

This study successfully proposes a dual-path framework combining DAF-Net and TDA for deep fusion of EEG and MEG:

- ✅ DAF-Net achieves 84% classification accuracy
- ✅ TDA provides 75-78% baseline performance
- ✅ Two paths capture complementary information effectively

This work provides new computational insights and technical solutions for multi-modal neural signal fusion.

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